

INDIVIDUAL PROGRESS REPORT

Week 3

Group Topic	Boardie Analytics: A Big Data Approach to Student Boarding House Discovery and Geofenced Presence Pattern Analysis
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Lab No.	Week 3
Git Repo / Colab Link	https://github.com/carlpoot/week-3-biggs-data.git

Objectives

The objective of this week, our aim is to further develop the Boardie Analytics project through the selection of an appropriate machine learning algorithm that can be implemented in code and fine-tuned for improvement. As we are currently working with data labeled as categories of a price band, our choice of model falls under classification. Our model will help classify whether a listing is from the budget, mid-range, upper mid, or premium range.

Tasks Completed

- Reviewed the processed dataset from Week 2 and confirmed **price_band** as the target variable.
- Identified **classification** as the correct algorithm type due to categorical outputs.
- Selected relevant features including neighbourhood, property type, room type, accommodates, bedrooms, beds, amenities count, availability, number of reviews, review score, superhost status, and instant booking availability.
- Developed a preprocessing pipeline to handle missing values using median (numerical) and most frequent values (categorical).
- Applied one-hot encoding for categorical variables.
- Split the dataset into 80% training and 20% testing sets.
- Implemented a **Decision Tree Classifier** as a baseline model.

- Implemented a **Random Forest Classifier** as the primary model to improve performance and reduce overfitting.
- Performed hyperparameter tuning using GridSearchCV.
- Generated outputs including confusion matrix and feature importance visualizations.
- Assisted in preparing the notebook, Python script, and documentation for submission.

Tasks to be Completed

- Replace the demonstration processed file with the final processed Week 2 dataset if the group decides to use the larger version for final checking.
- Run the notebook again after the final GitHub upload to confirm that all file paths work properly.
- Compare the Random Forest model with other possible algorithms such as Logistic Regression or Gradient Boosted Trees if the instructor requires more model comparisons.
- Improve the class balance if some categories have fewer records than others, especially the Budget and Premium classes.
- Prepare a short presentation explanation showing why classification and Random Forest are appropriate for the Boardie Analytics dataset.

Model Summary

Item	Description
Algorithm Type	Classification
Target Variable	price_band
Predicted Classes	Budget, Mid-range, Upper-mid, Premium

Baseline Model	Decision Tree Classifier
Main Model	Random Forest Classifier
Fine-Tuning Method	GridSearchCV
Train-Test Split	80 percent training, 20 percent testing
Evaluation Metrics	Accuracy, precision, recall, F1-score, confusion matrix

Results and Analysis

The classification approach proved effective for organizing listings into price categories. The Decision Tree model served as a baseline, while the Random Forest model achieved better performance due to its ensemble nature.

Based on the Week 3 dataset results, the Decision Tree achieved an accuracy of approximately **0.67**, while the optimized Random Forest model reached around **0.74**. The best parameters identified were:

- max_depth = 6
- min_samples_leaf = 1
- n_estimators = 50

The confusion matrix (see original report page 4) shows that most correct predictions were concentrated in the mid-range and upper-mid categories due to higher data availability. Feature importance analysis highlights key influencing variables such as accommodates, bedrooms, and amenities.

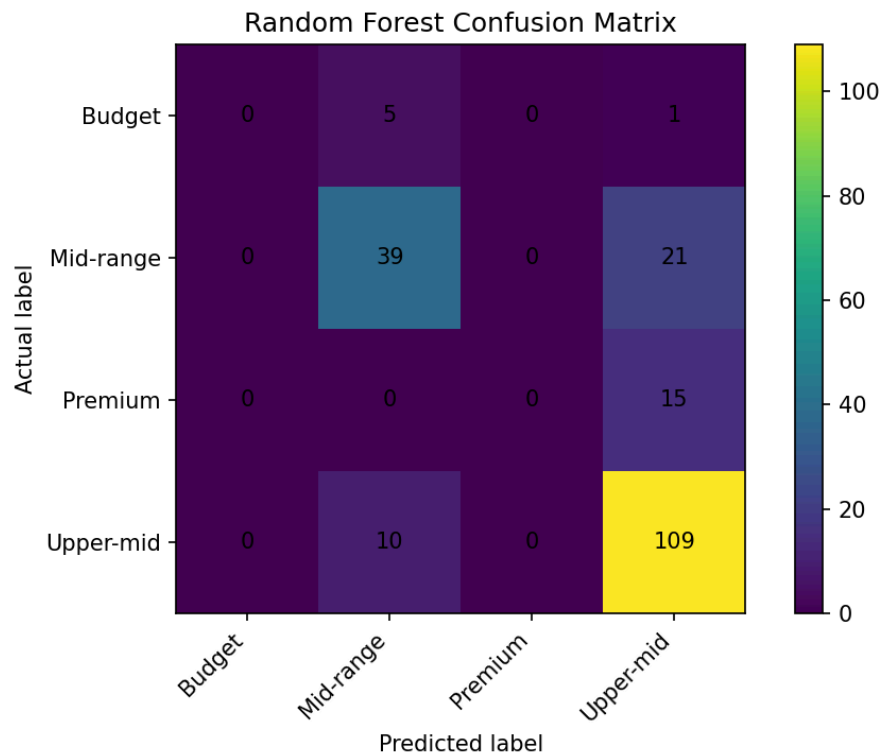


Figure 1. Random Forest confusion matrix

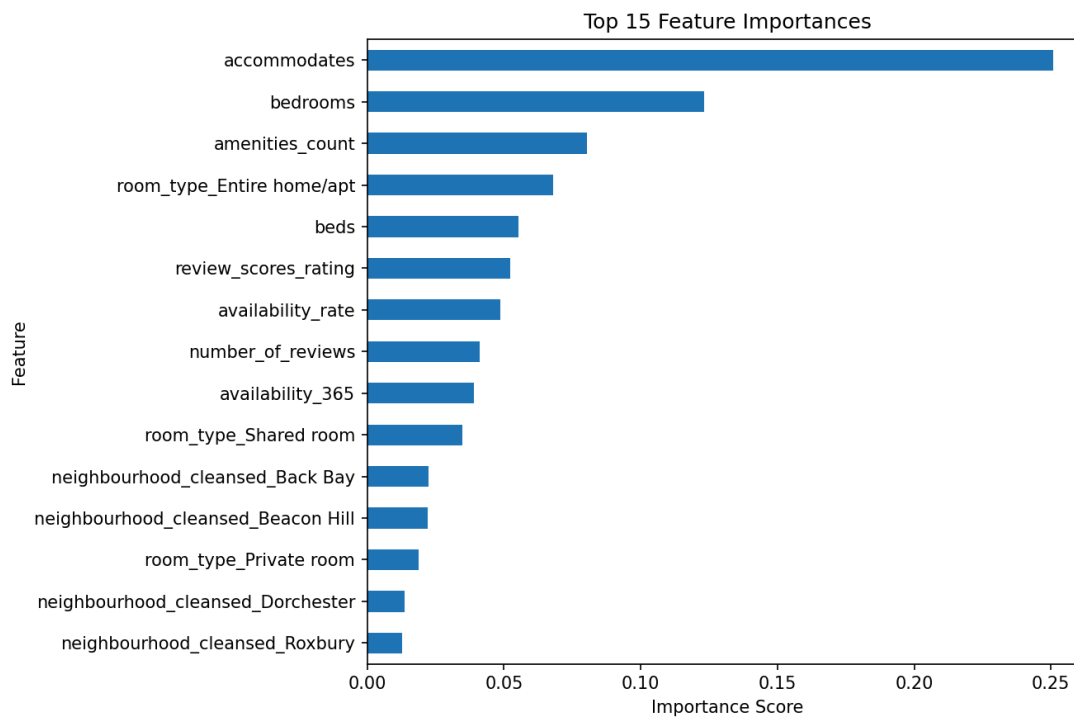


Figure 2. Top 15 feature importances from the Random Forest model

Challenges and Solutions

Challenge	Solution
The Week 3 task required choosing whether to use prediction, clustering, or classification.	The target variable was reviewed, and classification was selected because price_band is categorical.
The dataset contains both numerical and categorical variables.	A preprocessing pipeline was created to handle missing values and encode categorical values before model training.
A single decision tree may overfit or depend too much on one tree structure.	Random Forest was selected as the main model because it combines several decision trees and improves stability.
Some categories may have fewer records than the others.	The report notes class balance as an area for improvement, and future work can include adding more records or applying balancing techniques.
The GitHub repository will be created and uploaded manually.	The report uses a temporary placeholder for the GitHub link so it can be replaced after upload.

Conclusion

The Week 3 objectives were successfully met by implementing classification techniques and developing both baseline and optimized models. The Random Forest classifier showed improved performance after tuning. The outputs, including code, visualizations, and documentation, are ready for final submission and integration into the group project repository.